nature methods

SUPPLEMENTARY INFORMATION

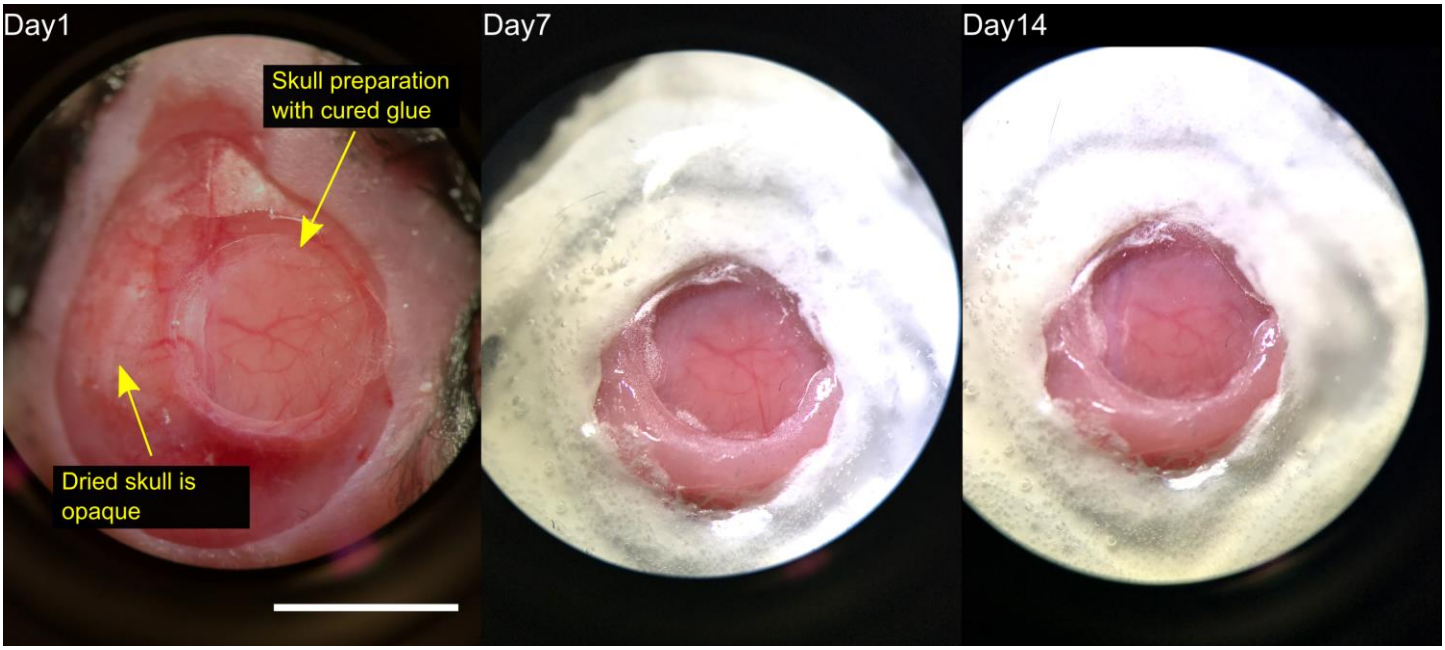
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Three-photon imaging of mouse brain structure and function through the intact skull

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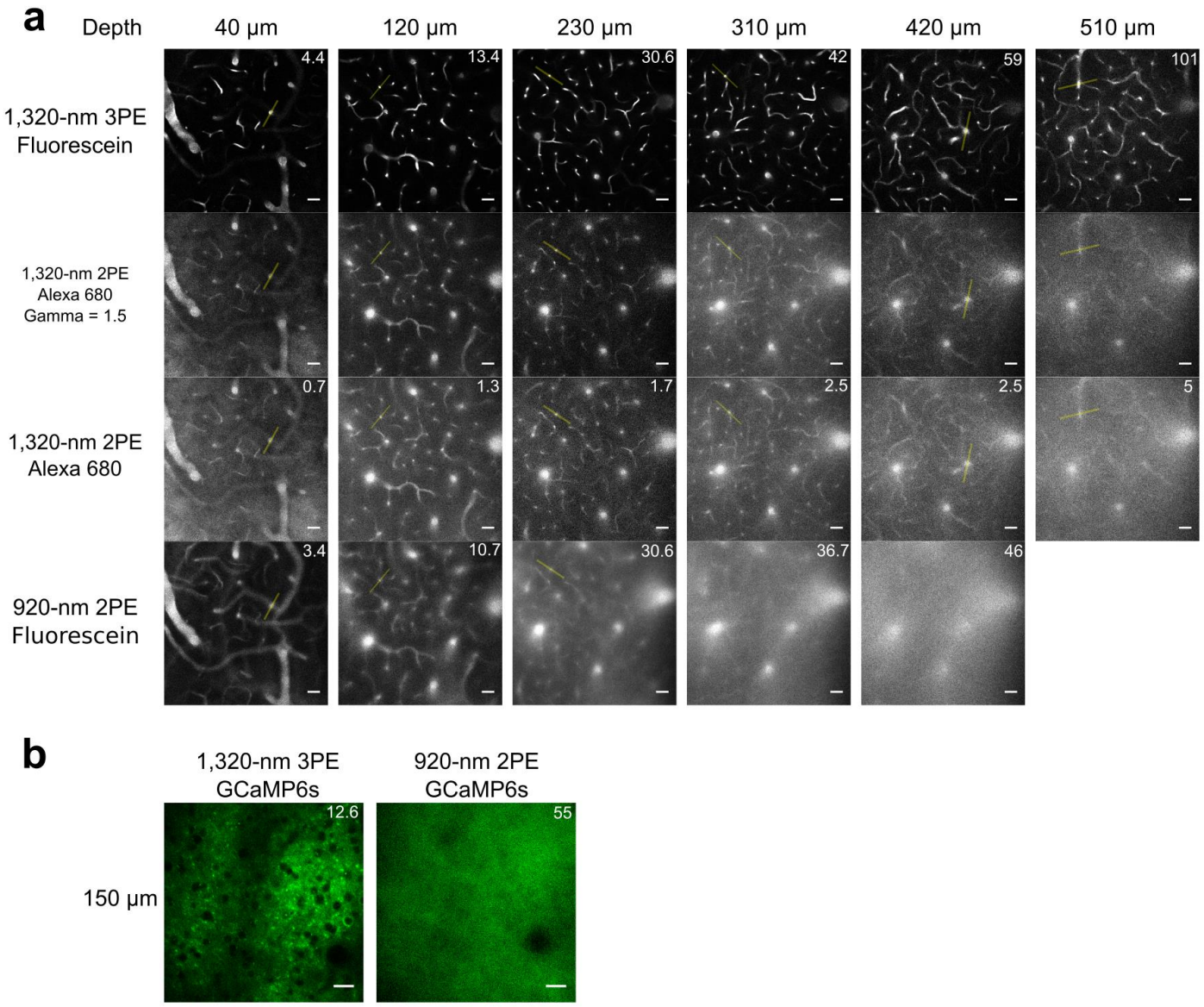
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Supplementary Figure 1

Visual examination of skull transparency after skull preparation.

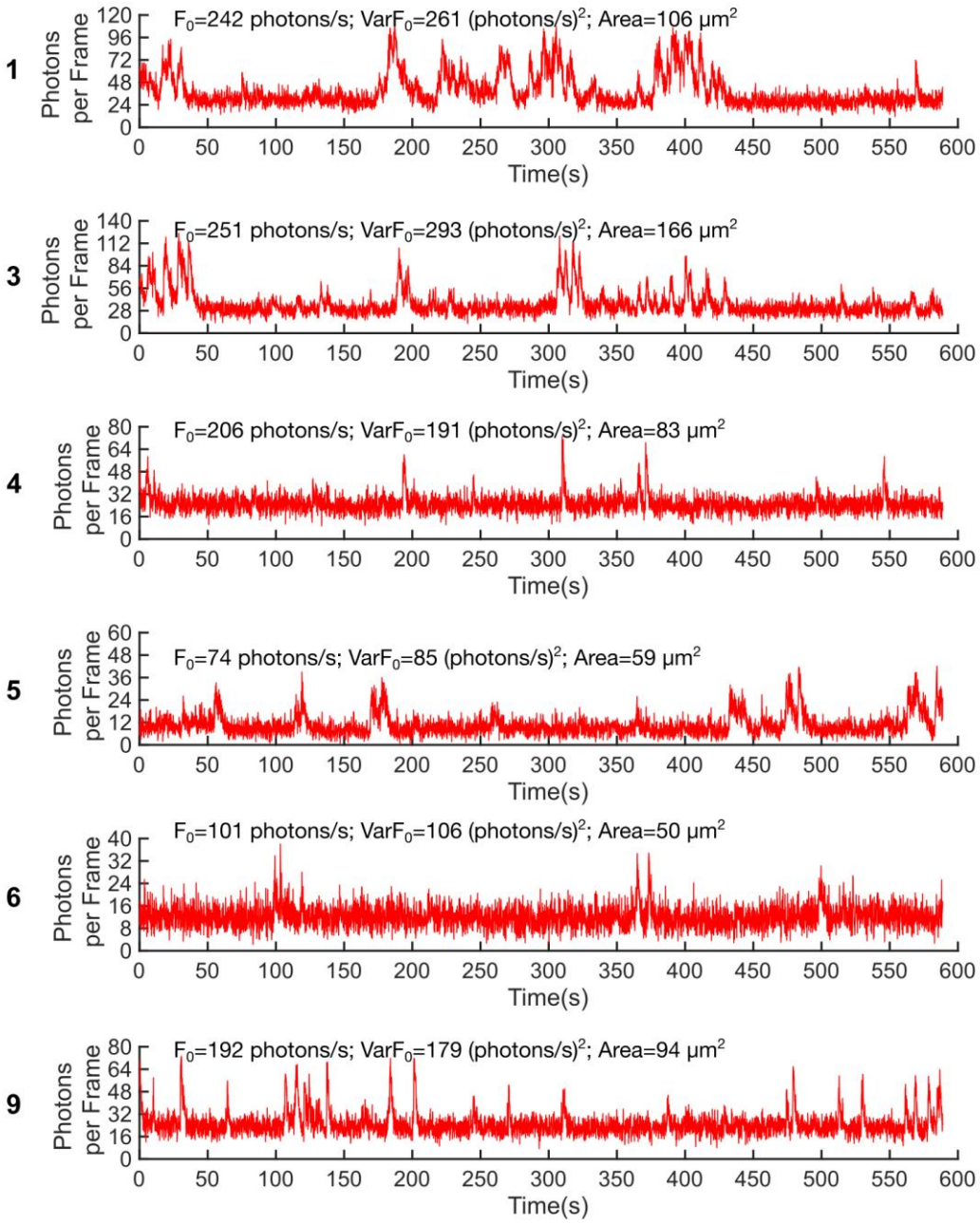
The bone surface was covered by Loctite 4305, and a #1 round coverslip (5 mm in diameter) was laid on top of it. The glue was cured by UV light. Day 1 was the day of the skull preparation. The images were taken under a stereoscope. Photos on day 7 and day 14 were taken after 2 and 3 recording sessions, respectively. Scale bar, 5 mm.



Supplementary Figure 2

Comparison of 1,320-nm 3PM, 1,320-nm 2PM, and 920-nm 2PM for vasculature imaging.

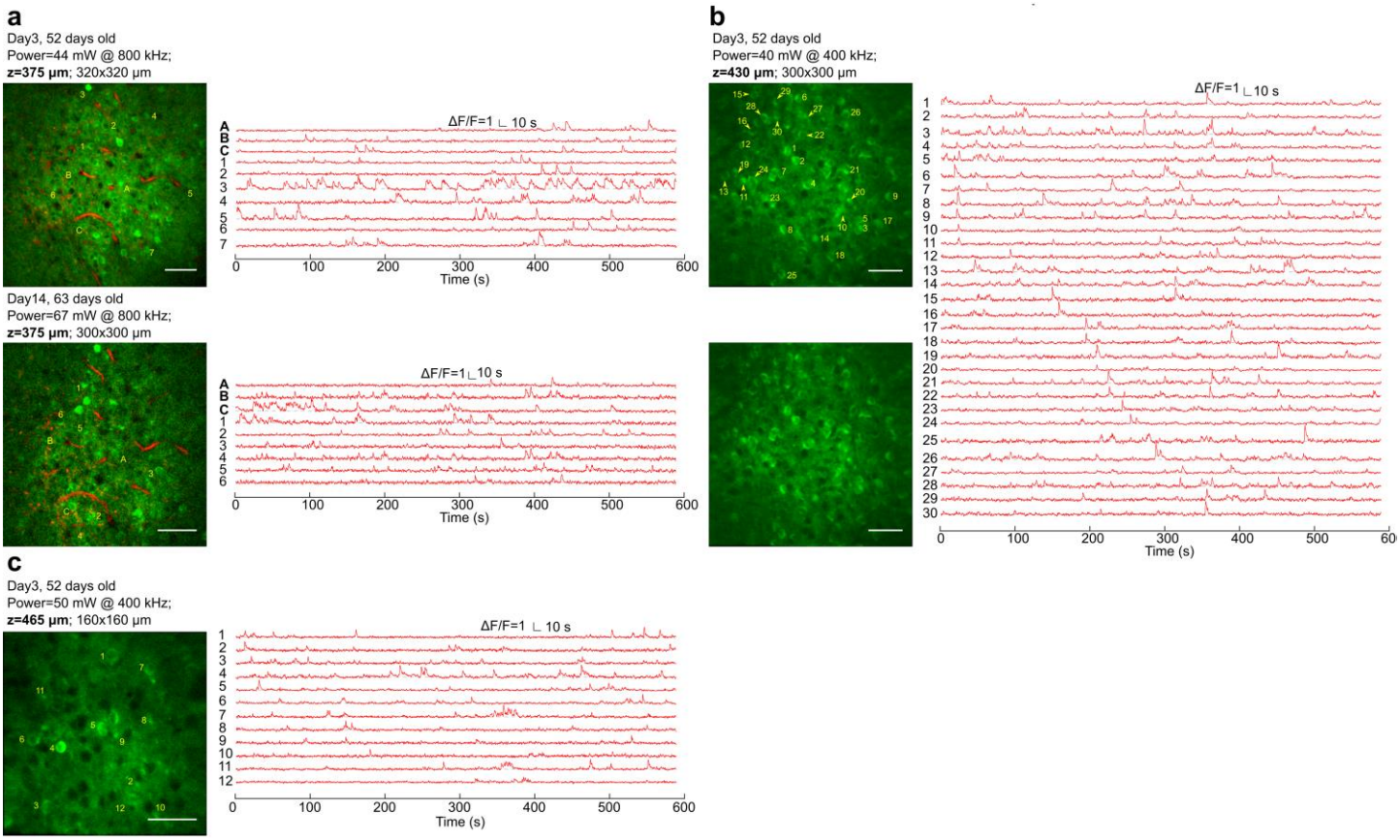
a, Vasculature imaging of a wild-type mouse (C57BL/6J, male, 12-week-old, similar results n=3) in the same cortical column under the central area of the parietal bone. Each site was imaged by 920-nm 2PE of fluorescein, 1320-nm 3PE of fluorescein, and 1320-nm 2PE of Alexa 680 during the same imaging session. The number in the upper right corner of each frame indicates the average power used in unit of mW. The 920-nm excitation source was operated at 80 MHz repetition rate, and 1320-nm at 400 kHz. Each frame was integrated for 50 s to ensure high signal-to-noise ratio. For the images in the row labeled as “1320-nm 2PE Alexa 680 Gamma=1.5”, each pixel value S is raised to $S^{3/2}$ to show that the lower contrast in 2PM images cannot be compensated by nonlinear contrast stretching as post-processing. For fair visual comparison between the different imaging modalities, all image contrasts are linearly stretched to saturate the brightest 0.2% pixels. Scale bar, 30 μm.
b, Comparison of 920-nm 2PM and 1320-nm 3PM through-skull imaging of neurons in a transgenic mouse (*CamKII-tTA/tetO-GCaMP6s*, female, 10-week-old, n=1) at 150 μm cortical depth with ~ 105 μm skull thickness. Each frame was integrated for 40 s. Scale bar, 20 μm.



Supplementary Figure 3

Raw data of through-skull neuronal activity traces.

The unprocessed version of a subset of the activity traces in Fig. 2f, representing neurons of different size, brightness, and activity pattern (similar results n=5). The signal of each neuron (in photons/neuron/frame) was plotted against time, at the original 8.49 Hz recording frame rate, without any temporal down-sampling or low-pass filtering. Fluorescence intensity was converted to photon counts per frame per neuron to confirm that the baseline fluctuations are mostly due to photon shot noise. Above each trace, the value of the measured mean (F_0) and variance ($VarF_0$) of the baseline fluorescence photon-count rate, and the areas of each neuron, are indicated. For each trace, its processed version and the corresponding neuron can be found in Figs 2e and 2f respectively, by its index to the left of the trace.



Supplementary Figure 4

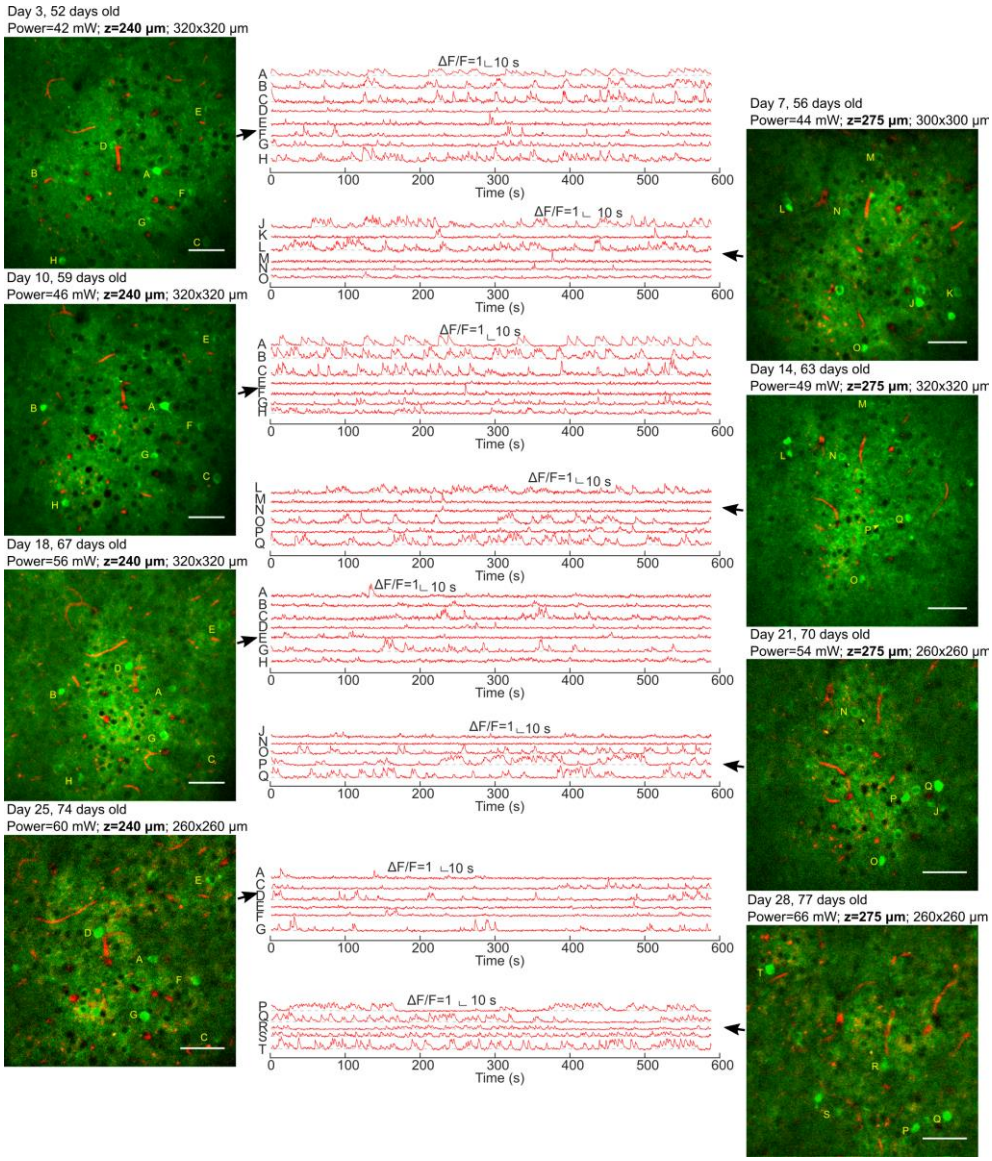
Through-skull recording of neuronal activity in deeper cortex in an awake animal.

Spontaneous activity traces from deeper cortical neurons recorded in a GCaMP6s mouse under awake condition (*CamKII-tTA/tetO-GCaMP6s*, female, 8-9 weeks old, n=1). In each panel, the high resolution image on the left shows the recording site (green: GCaMP6s fluorescence; red: THG; Scale bar, 50 ⌈m). The neurons are indexed for their corresponding time traces. The day since the skull preparation, the age of the mouse, the average power, the repetition rate, the imaging depth, and the size of FOV are indicated at the top of each image. The right panels show activity traces of the neurons, recorded at 8.49 Hz frame rate. The traces were normalized to their baselines and low-pass filtered by a hamming window with 1.06 s time constant. The thickness of the skull was about 120 μm.

a, A recording site located at 375 μm cortical depth observed in two imaging sessions 11 days apart (similar results n=3). We leveraged the THG channel to visualize the vasculature map, which serves as the landmark for locating the same recording site. Due to small variations in each imaging session (e.g., mouse orientation, tilting, the finite focal depth, etc.), however, it is difficult to match the recording site exactly. Nonetheless, some neurons can be identified in both imaging sessions. These neurons are indexed alphabetically, and the same labels are used for both imaging sessions. Neurons only observed in one of the two imaging sessions are indexed numerically.

b, A recording site located at 430 μm cortical depth (n=1). An additional image without any neuron indexing (unlabeled) is shown for better visualization of the image.

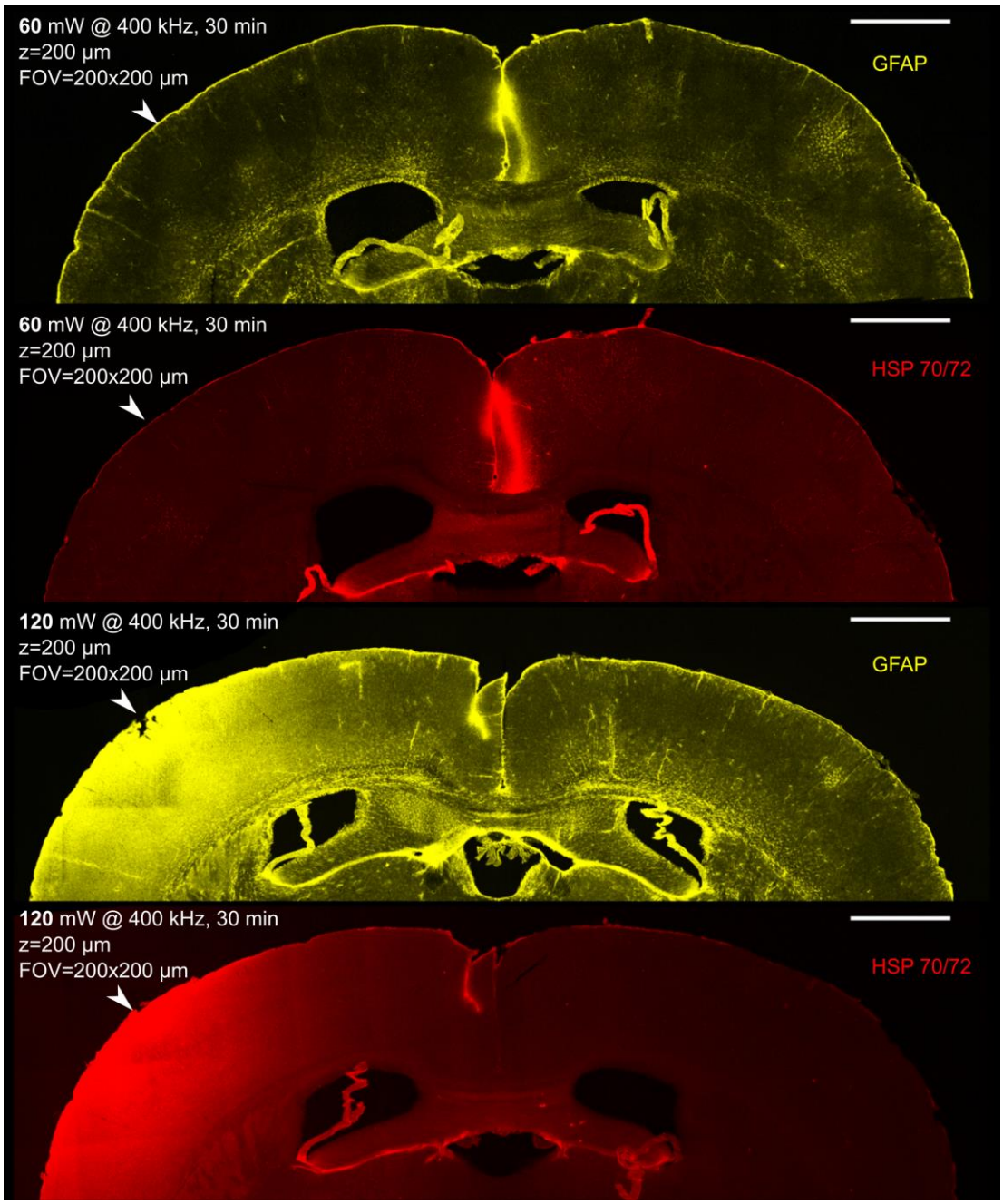
c, The deepest recording site in this study located at 465 μm cortical depth (n=1).



Supplementary Figure 5

Longitudinal through-skull recording of neuronal activity at the same locations in an awake animal over 4 weeks.

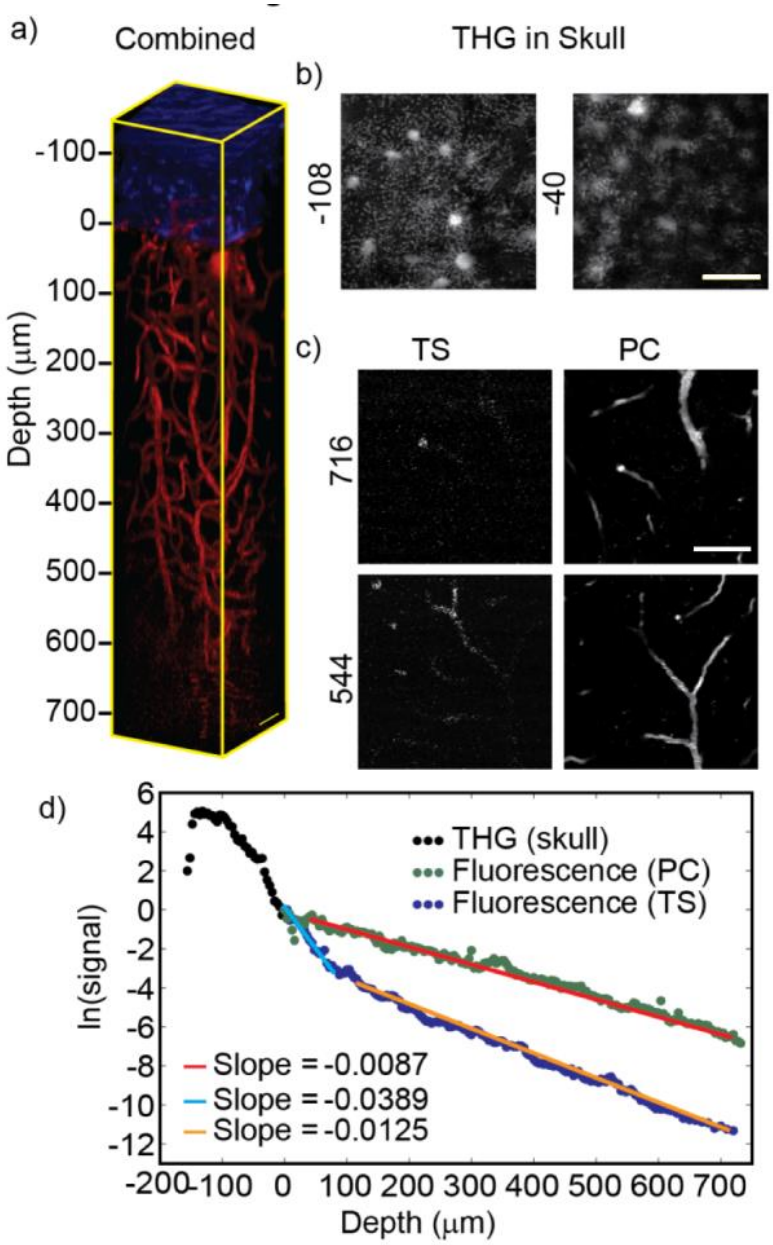
Spontaneous activity traces from cortical L2/3 neurons were recorded in a GCaMP6s mouse under awake condition (*CamKII-tTA/tetO-GCaMP6s*, female, 8-11 weeks old, and similar results n=3) over a period of 4 weeks after the skull preparation. The left panels show high resolution images of a recording site at 240 μm cortical depth, and the right panels show another site at 275 μm depth (the same site as in Fig. 2e and 2f). Green: GCaMP6s fluorescence; red: THG; Scale bar, 50 μm. THG visualizes the vasculature map, which was used to navigate to approximately the same location for recording. As discussed in Supplementary Fig. 4, it is difficult to match the recording site exactly for each imaging session. Therefore, only a fraction of the neurons can be identified in multiple imaging sessions. These neurons are indexed alphabetically, and the same unique letter for each neuron is used in all imaging sessions. The day since the skull preparation, the age of the mouse, the average power used for activity recording, the imaging depth, and the size of FOV are indicated at the top of each image. 8 recording sessions were performed at each site (4 at each site are shown). The middle panels show the activity traces of the indexed neurons, recorded at 8.49 Hz frame rate, and each was normalized to its baseline and low-pass filtered by a hamming window with 1.06 s time constant. The repetition rate for all imaging was 800 kHz. The thickness of the skull was about 120 μm.



Supplementary Figure 6

Immunolabeled coronal sections of mouse brains after through-skull imaging at 1,320 nm.

The brain was imaged by 1320-nm 3PM operating at 400 kHz repetition rate. The imaging depth was about 200 μm under the cortical surface, and the scanned area was 200x200 μm, at the location of ~2.5 mm lateral and 1 mm caudal to the bregma point. The illumination continued for 30 minutes, and the brain was fixed after 16 hours (n=1). Scale bar, 1 mm.



Supplementary Figure 7

Through-skull imaging of mouse vasculature with 1,700-nm 3PM.

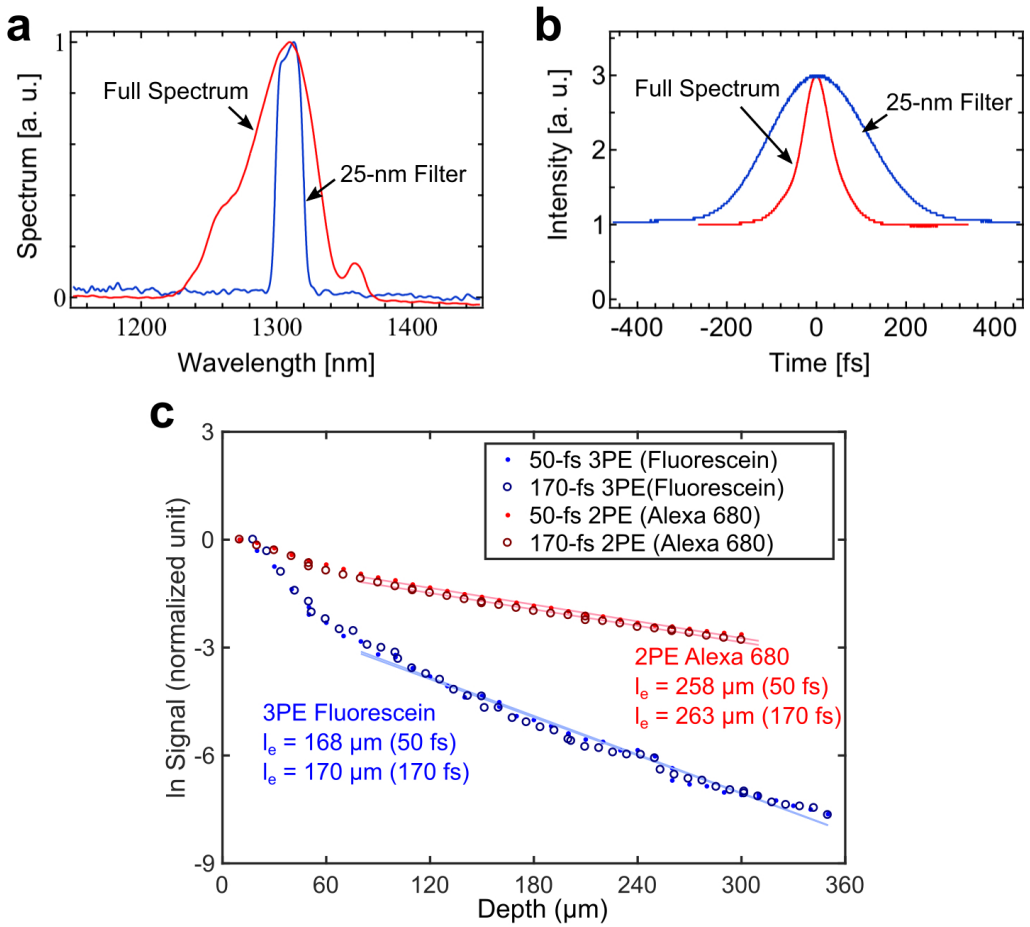
a, Through-skull imaging of Texas Red labeled vasculature of mouse brain with an acute preparation using water to fill the space between the cover glass and the skull (red, fluorescence; violet, THG, similar results n=3). The skull-brain interface is indicated by the boundary between the fluorescence and the bright THG signal.
b, THG images of osteocytes within the mouse skull (n=3).
c, Comparison of through-skull (TS) and post-craniotomy (PC) frames for the same mouse and FOV at two depths below the brain surface (n=1).
d, Signal attenuation before and after craniotomy (n=1). The plot displays THG signal attenuation within the skull (black dots), fluorescence signal attenuation within the brain below the skull (blue dots), and fluorescence signal attenuation in the same region of the brain post-craniotomy (green dots). The curves were matched at depth = 0 (i.e., the skull-brain interface). The attenuation length of the brain, through-skull, was 240 μm, and the attenuation length post-craniotomy was 345 μm. Scale bars, 50 μm.

Supplementary Note 1:
Detected fluorescence vs imaging depth for through-skull 2PM and 3PM at 1320 nm

Multiphoton excited fluorescence in the mouse brain can be separated into two parts: the fluorescence within the focal volume (i.e., the signal) and the fluorescence outside the focal volume (i.e., the background). We emphasize that this background includes all fluorescence excitation away from the focal volume, not only in the axial dimension but also in the transverse dimension. The detected fluorescence is proportional to the sum of the signal and the background, while the image contrast is determined by the signal-to-background ratio (SBR). It was shown before that the in-focus signal is generated by ballistic photons, and both ballistic and scattered photons generate the out-of-focus background[1]. With increasing imaging depth, the signal decreases exponentially due to the attenuation of the ballistic photons, and the relative contribution of the background to the detected fluorescence increases. While the signal in 2PM and 3PM behave similarly with the same excitation wavelength and same mouse brain, the out-of-focus background is much more pronounced in 2PM than in 3PM, as shown in Fig. 1. In fact, for through-skull 2PM, the background contribution to the detected fluorescence is significant even at shallow depth (Fig. 1). Because the background fluorescence does not decrease rapidly with imaging depth, this leads to the apparent reduction in attenuation of the detected fluorescence for through-skull 2PM. For example, the measured effective attenuation length[2] (l_e) for through-skull 2PM (l_e=230 μm) is longer than that of 3PM (l_e=170 μm) (Fig. 1e), which is indicative of the unwanted background contribution. Indeed, the out-of-focus background in through-skull 2PM is responsible for the poor imaging contrast even at shallow depth below the skull.

The measured effective attenuation length (l_e=170 μm, Fig. 1e) in the neocortex for through-skull 3PM is much shorter than that measured with cranial windows (approximately 300 μm throughout the entire neocortex[2]). Such rapid signal attenuation can be explained by the deterioration of the PSF in through-skull imaging. As the imaging depth increases, the beam size at the skull surface also increases, which can lead to significant aberration due to the heterogeneity of the skull. As the PSF deteriorates with increasing imaging depth, less signal is generated from the focal volume and more background is produced in the wings of the PSF, even with 3PE. This is corroborated by the decline in the SBR for through-skull 3PM (Fig. 1d), which has an SBR ~ 10 at the imaging depth of ~ 500 μm. Previous work of 3PM with cranial windows achieved SBR ~ 100 at > 1000 μm imaging depth[3]. The measured effective attenuation length for through-skull imaging represents the combined effect of scattering by the brain tissue and wavefront distortion by the skull. Therefore, the effective attenuation length of the brain tissue in through-skull 3PM is much shorter than that with imaging through the cranial window.

We have neglected the dependence of fluorescence collection efficiency on imaging depth in the above analysis, which is justified because of the relatively shallow imaging depths (< 500 μm) in this study. Furthermore, the different pulse width used in 2PM and 3PM has no measurable impact on the data obtained in Fig. 1e. (see Supplementary Fig. SN1. 1).



Supplementary Note 1 Figure 1 | Tissue Attenuation Length vs Excitation Pulse Duration. **a**, The full spectrum of the excitation pulses and the spectrum filtered by a 25-nm bandpass filter (a.u. means arbitrary units; n=3).
b, Intensity autocorrelation traces of the original pulses and the spectrally filtered pulses. The measurements were performed under the objective, and the pulse durations were measured to be 50 fs and 170 fs, respectively, assuming sech^2 pulse shape (a.u. means arbitrary units; n=3).
c, The effective attenuation lengths measured through an intact skull of ~120 μm thickness (n=2). The experiment setup was similar to that used for Fig. 1e, except that pulses of 50 fs and 170 fs were used to measure the attenuation lengths. The effective attenuation lengths were measured for 3PE of fluorescein and 2PE of Alexa 680. Despite the large difference in pulse duration, the difference in the effective attenuation lengths is negligible.

Supplementary Note 2:
Tissue Damage Assessment using Immunohistology and the limits of through-skull 3PM

The immunostaining results (Supplementary Fig. 6) show that 60 mW average power under the objective lens at 400 kHz repetition rate did not cause measurable damage, while 120 mW at 400 kHz repetition rate caused activation of both heat shock protein and glial fibrillary acidic protein. These results are consistent with our longitudinal studies, where neurons remained viable after hours of exposure to as high as ~ 70 mW under the objective lens at 800 kHz repetition rate. The thermal damage threshold for through-skull imaging where the skull is intact is probably lower than that for imaging through the cranial window where a significant portion of the skull is removed. As shown in one previous study, the high thermal conduction through the cranial window plays a significant role in cooling the brain surface, and the maximum temperature rise occurred at hundreds of microns below the brain surface when imaging through the cranial window[4].

No measurable nonlinear tissue damage was detected in the imaging plane (~ 200 μm below the brain surface) with 60 mW average power at 400 kHz repetition rate (Supplementary Fig. 6). The activity recordings at depths between 240 μm and 400 μm used similar average power at 800 kHz repetition rate, and had larger FOV (Fig. 2, Supplementary Figures 3 and 4). Therefore, no measurable nonlinear photodamage is expected in the focal plane under the conditions for activity imaging in this study.

The limits on FOV, temporal resolution, and imaging depth for activity recording are ultimately determined by the characteristics of the fluorescent probe, and the maximum allowable average power under the objective lens and peak power at the focus. In shot-noise-limited optical detection and assuming that the sampling rate is sufficiently high relative to the response time of the indicator, the quality of the activity recording can be quantified by the discriminability parameter[5], d', which is determined by the mean value of the baseline fluorescence photon-count rate per neuron (F_0, Supplementary Figures 3) and the characteristics of the indicator, i.e., the ΔF/F and the *1/e* decay time τ, $d' = (\Delta F/F)\sqrt{F_0\tau/2}$ (See ref. 2 for details). Using the values for GCaMP6s (ΔF/F for a single action potential ~ 30% and τ ~ 2.0 s)[6], F_0 ~ 100 photons/neuron/s is required for d'=3, which yields ~ 93% true detection and 7% false detection rate for an isolated single action potential[5]. If d' ~ 3 is used as the criteria for acceptable recording quality, our immunostaining results and chronic imaging results indicate that a 320 μm x 320 μm FOV in the cortical layer 2/3 is achievable for through-skull 3PM of GCaMP6s without measurable tissue damage. Deeper imaging is possible by reducing the repetition rate of the excitation (e.g., 400 kHz, Supplementary Figure 4c), albeit at the expense of a reduced FOV or frame rate.

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